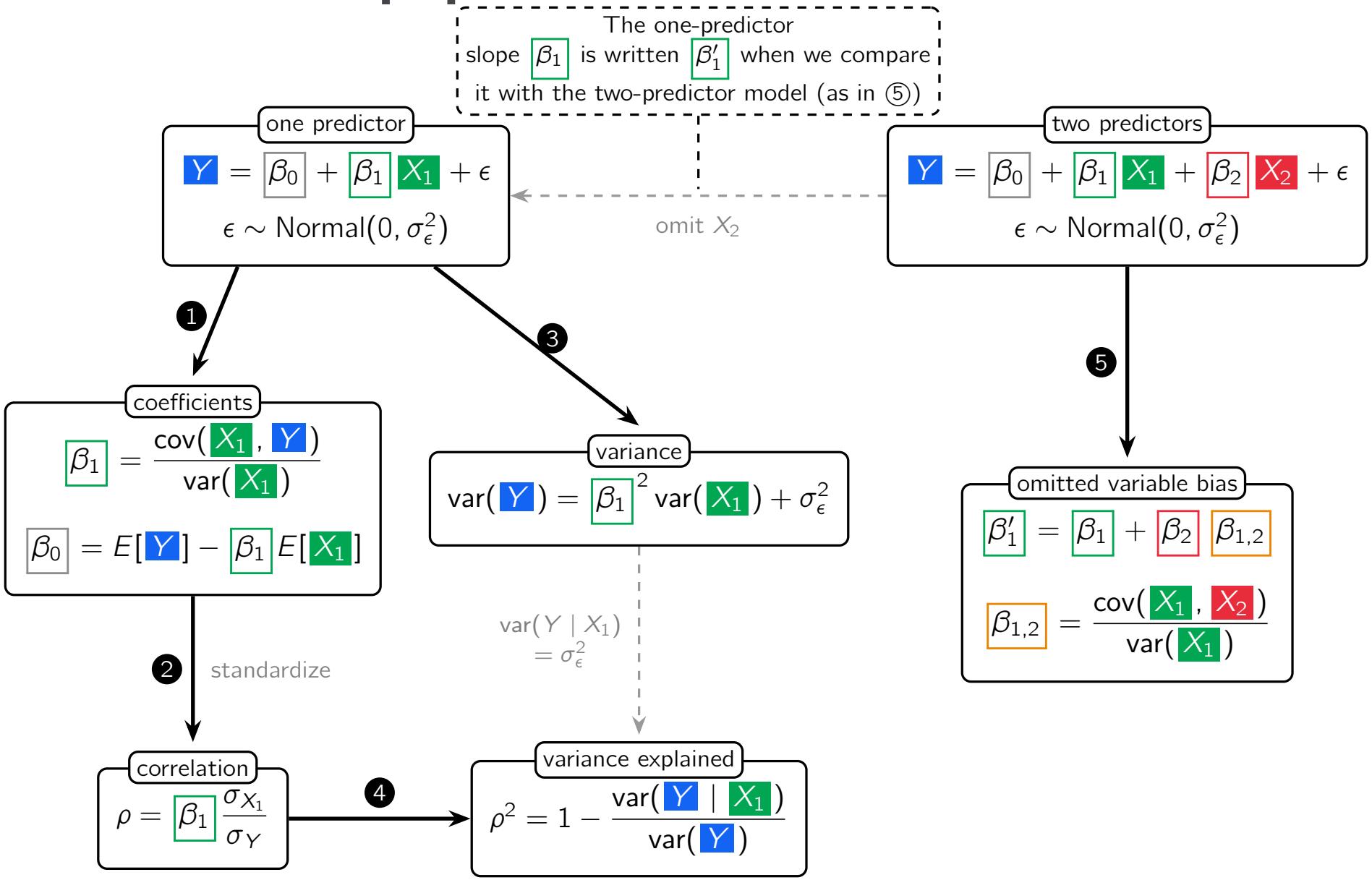
Lecture 18 (10/26)

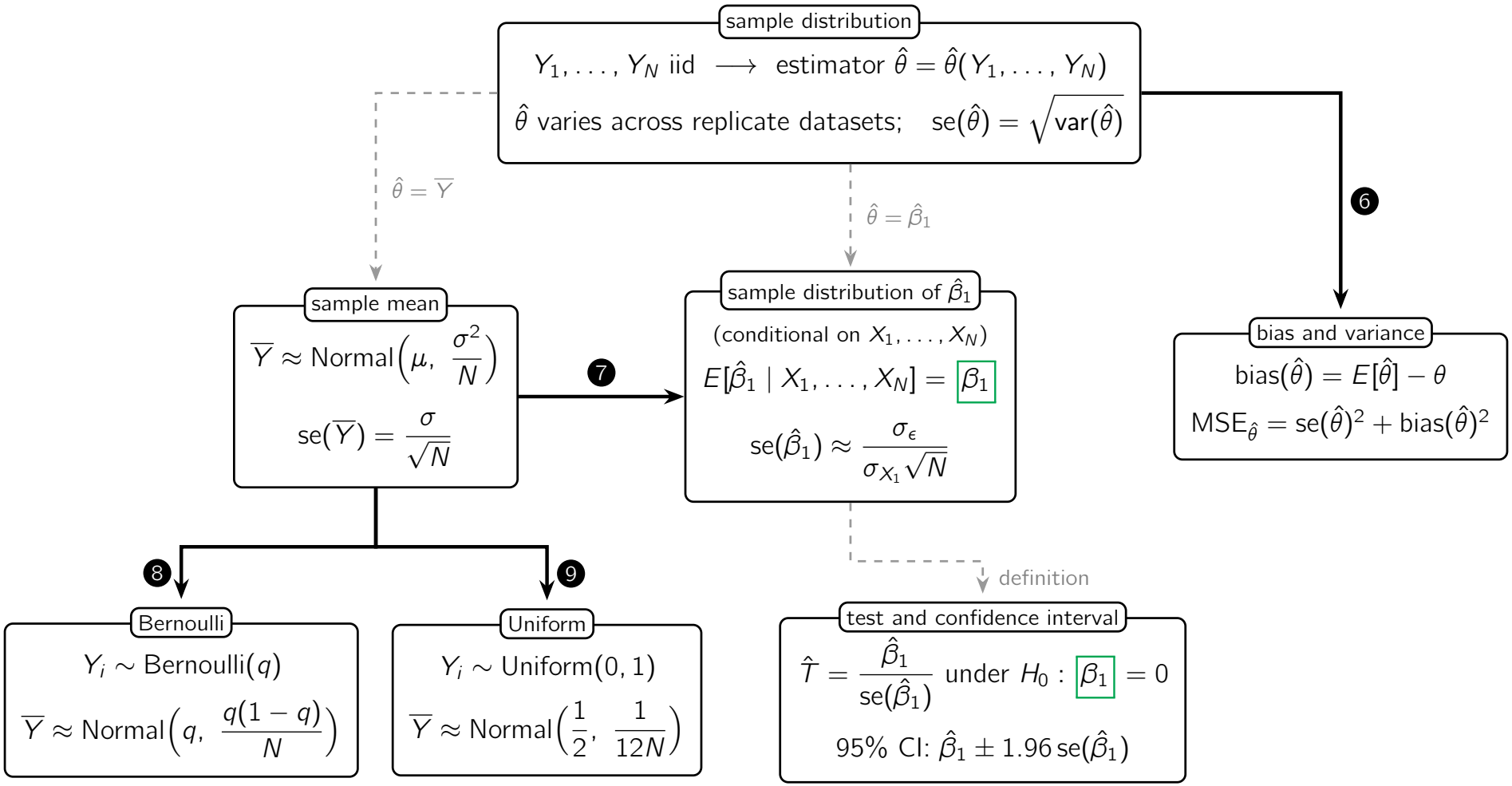
Review for Midterm 2

Derive-athon: population formulas



Unit notes: ① §2.3 the slope ② §2.3 regression to the mean ③ §2.4 coeff. of determination ④ §2.4 correlation ⑤ §4.2 omitted variable bias

Derive-athon: estimators



Unit notes: 6 §3.2 bias-variance tradeoff 7 §3.4 sample distribution of $\hat{\beta}_1$ 8 §2.1, §3.1 CLT 9 §1.4, §3.1; $\text{var}(Y_i) = \frac{1}{12}$ in the 9/25 slides

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9) **False.** If $X_2 = X_1^2$, then $E[Y \mid X_1] = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2$ is not linear in X_1. It holds if $X_2 = a + \beta_{1,2}X_1 + \eta$ with η Normal and independent of X_1 (noise $\beta_2\eta + \epsilon$).

Switching the axes

Suppose $Y = \beta_0 + \beta_1 X + \epsilon$, with ϵ independent of X, and let ρ be the correlation between X and Y. Now fit the single-predictor model with X as the response and Y as the predictor:

$$X = \gamma_0 + \gamma_1 Y + \eta.$$

Question

Write γ_1 in terms of β_1 and ρ. When is $\gamma_1 = 1/\beta_1$?

See also Problem 2.5 (swapping response and predictor variables) in the Unit 2 notes.